

Tissue	GSE	Samples	Genes	Classes	Shannon entropy
Breast	GSE71651	33 [15 tumor, 18 normal]	58735	2	0.994

GSE71651 - With the 10 Top Genes & Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.626±0.025	0.594±0.016	1.738±0.235	0.620±0.045	0.600±0.000	0.649±0.046
SVM	0.743±0.042	0.722±0.051	2.989±0.618	0.351±0.105	0.740±0.074	0.745±0.040
NB	0.697±0.000	0.750±0.000	1.799±0.000	0.000±0.000	1.000±0.000	0.444±0.000
KNN	0.636±0.000	0.500±0.000	2.395±0.000	0.720±0.000	0.400±0.000	0.833±0.000
LR	0.727±0.000	0.727±0.000	2.402±0.000	0.300±0.000	0.800±0.000	0.667±0.000
MLP	0.639±0.069	0.600±0.096	1.908±0.626	0.594±0.211	0.609±0.135	0.665±0.078
RF	0.755±0.033	0.710±0.043	4.208±1.766	0.407±0.071	0.662±0.057	0.831±0.033
DUM	0.545±0.000	0.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000	1.000±0.000

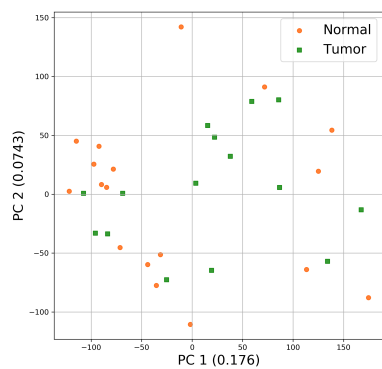
GSE71651 - With all Genes & Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.549±0.052	0.477±0.051	1.276±0.300	0.890±0.202	0.452±0.059	0.631±0.083
SVM	0.449±0.034	0.105±0.026	0.325±0.144	1.223±0.097	0.071±0.017	0.764±0.055
NB	0.273±0.000	0.250±0.000	0.370±0.000	2.637±0.000	0.267±0.000	0.278±0.000
KNN	0.606±0.000	0.552±0.000	1.601±0.000	0.700±0.000	0.533±0.000	0.667±0.000
LR	0.485±0.000	0.414±0.000	0.901±0.000	1.079±0.000	0.400±0.000	0.556±0.000
MLP	0.485±0.073	0.343±0.097	0.877±0.390	1.125±0.240	0.301±0.099	0.638±0.114
RF	0.438±0.033	0.268±0.072	0.590±0.168	1.267±0.126	0.231±0.074	0.609±0.054
DUM	0.545±0.000	0.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000	1.000±0.000

GSE71651 - With all Genes & without Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.549±0.050	0.475±0.061	1.278±0.368	0.881±0.148	0.449±0.069	0.633±0.070
SVM	0.545±0.000	0.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000	1.000±0.000
NB	0.273±0.000	0.250±0.000	0.370±0.000	2.637±0.000	0.267±0.000	0.278±0.000
KNN	0.576±0.000	0.500±0.000	1.402±0.000	0.799±0.000	0.467±0.000	0.667±0.000
LR	0.485±0.000	0.414±0.000	0.901±0.000	1.079±0.000	0.400±0.000	0.556±0.000
MLP	0.559±0.073	0.509±0.090	1.379±0.512	0.840±0.235	0.510±0.116	0.600±0.113
RF	0.457±0.041	0.334±0.071	0.735±0.192	1.197±0.160	0.303±0.079	0.584±0.040
DUM	0.545±0.000	0.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000	1.000±0.000

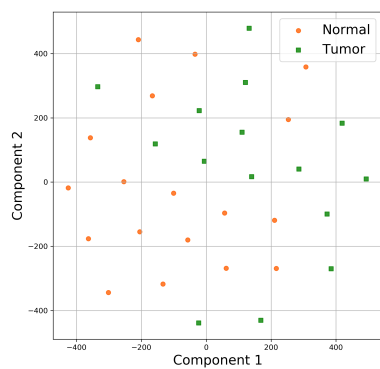
GSE71651 - With the 10 Top Genes & Hyperparameter Optimization & Class weight						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.712±0.029	0.640±0.048	3.370±0.402	0.520±0.077	0.568±0.064	0.831±0.010
SVM	0.480±0.046	0.432±0.061	0.915±0.187	1.106±0.204	0.439±0.078	0.514±0.061
LR	0.515±0.000	0.500±0.000	1.066±0.000	0.934±0.000	0.533±0.000	0.500±0.000
RF	0.536±0.061	0.460±0.068	1.230±0.477	0.932±0.188	0.437±0.071	0.618±0.089

GSE71651 - With the 10 Top Genes & Hyperparameter Optimization & SMOTE						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.665±0.017	0.671±0.018	1.941±0.153	0.489±0.043	0.685±0.026	0.645±0.028
SVM	0.710±0.036	0.717±0.036	2.389±0.442	0.388±0.081	0.737±0.052	0.683±0.054
NB	0.722±0.000	0.783±0.000	1.799±0.000	0.000±0.000	1.000±0.000	0.444±0.000
KNN	0.694±0.000	0.703±0.000	2.168±0.000	0.417±0.000	0.722±0.000	0.667±0.000
LR	0.694±0.000	0.703±0.000	2.168±0.000	0.417±0.000	0.722±0.000	0.667±0.000
MLP	0.712±0.066	0.722±0.070	2.423±0.785	0.372±0.157	0.756±0.098	0.667±0.079
RF	0.779±0.030	0.776±0.029	3.837±0.760	0.297±0.053	0.765±0.040	0.792±0.047
DUM	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000

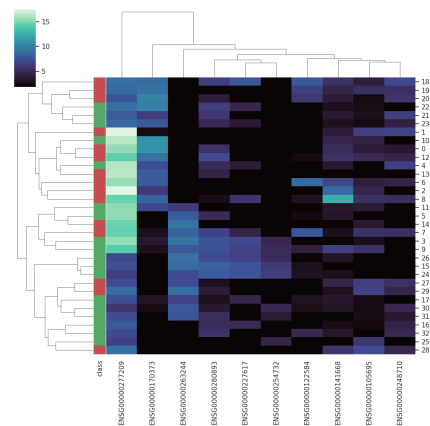
<https://sbc.b.inf.ufrgs.br/barracurda>



(a) PCA GSE71651



(b) t-SNE GSE71651



(c) HeatMap GSE71651 - Top 10 Genes

<https://sbc.inf.ufrgs.br/barracurda>